

Curriculum Vitae

Personal Data

Patrick Wilhelm



Contact Berlin
Mobile: +49 176 41785493
E-Mail: patrick.wilhelm@tu-berlin.de
Date of Birth 08.02.1997
Professional Profiles  Google Scholar  LinkedIn

Research Interests

AI Safety **Chain-of-Thought Monitoring and Reward Hacking**
ICML 2026 – *From Reward-Hack Activations to Agentic Risk States: Context-Calibrated Mechanistic Monitoring in LLM Agents*
Workshop on Agents in the Wild: Safety, Security and Beyond [Paperlink](#)
ICLR 2026 – *Monitoring Emergent Reward Hacking During Generation via Internal Activations*
Workshop on Principled Design for Trustworthy AI [Paperlink](#)

Sustainable Computing **Efficient and Sustainable AI Systems**
EuroMLSys 2025 – *Beyond Test-Time Compute Strategies: Advocating Energy-per-Token in LLM Inference* [Paperlink](#)
5th Workshop on Machine Learning and Systems, co-located with EuroSys 2025
HotCarbon 2025 – *Noise-Aware Client Selection for Carbon-Efficient Federated Learning via Gradient Norm Thresholding* [Paperlink](#)
Workshop on Sustainable Computer Systems

University

09.2022 - 02.2023 **Data Science** (Master of Science – Erasmus), Sapienza Università di Roma
Major in Deep Learning and Control of Energy Networks

10.2020 - 2023 **Computational Engineering Science** (Master of Science), Technische Universität Berlin
Focus in Machine Learning and Data Science – Final Grade: 1.6

10.2016 - 04.2020 **Mechanical Engineering** (Bachelor of Engineering)
University of Applied Sciences Bielefeld; Focus: Energy Engineering
Grade: 2.6

Practical Experience

- 03.2024 - today* **PhD – Research Associate**, TU Berlin, Distributed Operating Systems Lab (DOS)
- Funding and PhD Graduate Student at BIFOLD (one of six national AI competence centers in Germany) – Berlin Institute for the Foundations of Learning and Data
- Researching about Reasoning AI Models and Chain-of-Thought Monitoring Systems as well as Sustainable Computing
- Teaching and Supervising Students
- 05.2023 - 12.2023* **Master Thesis**, Siemens AG, Corporate Technology – Digital Manufacturing Technologies, Munich

- “Development of a Generative AI for Decision-Making in Printed Circuit Board Design” – Grade: 1.0
- Diffusion Models and Reinforcement Learning for Trajectory Planning
- **Impact:** Foundation for a personalized AI assistant for PCB designers based on Diffusion Models
- 04.2023 - 05.2023* **Internship**, Siemens AG, Corporate Technology – Digital Manufacturing Technologies, Munich
- NLP and Large Language Models for Product Recommendation
- 10.2022 - 04.2023* **Student Project**, Volkswagen AG, Wolfsburg/Kassel
- Quality Management in critical software development. Working with Automotive SPICE, ISO 26262 and KEEP.
- 04.2022 - 04.2023* **Battery Data Analyst, Working Student**, Vattenfall AB, Battery Innovation, Berlin
- Mathematical optimization with Gurobi
- Physical modelling of Battery-cell degradation
- 08.2020 - 04.2022* **Working Student**, Siemens AG, Smart Infrastructure, Berlin
Analytics of SAP data for service and sales using Python
- 10.2019 - 04.2020* **Intern and Bachelor Thesis**, thyssenkrupp AG Research Department
- Data-driven physical modeling of streaming machines
Bachelor Thesis:
- Development of data-driven soft sensors for process monitoring of flow processes; Grade: 1.7
Impact: Foundation for a predictive-maintenance system for the burning process within a cement plant

02.2019 - 10.2019 **Working Student**, DMG MORI AG
Software Solutions Department

Skills & Languages

Computing Systems:	<i>PyTorch Profiling, GPU Monitoring, vLLM/DeepSpeed, NVIDIA Man- brary – NVML</i>
Machine Learning:	<i>Federated Learning, Reinforcement Learning , etc.</i>
Languages:	<i>English, German, Italian (B1), Russian (A1)</i>
Project Development and Automotive:	<i>Scrum, Automotive SPICE, ISO 26262</i>

Berlin, July 16, 2026